

Descriptive Analytics Report: Steam

Game Sales, Ratings & Market Trends

Note on the topic:

Steam is a digital platform developed by Valve Corporation for **purchasing, downloading, and playing PC games**. It offers thousands of games across different genres and includes features like user reviews, multiplayer support, and cloud storage. Boasting over **100 million** active users globally.

Importance of Game Analytics

Game analytics helps developers and publishers understand player **preferences, game popularity, pricing trends, and user feedback**. It supports better decision-making in game development, marketing, and business planning. Monitoring these market trends ensures that gaming companies can strategically position their upcoming titles and adapt to changing player preferences.

Objective

Build a descriptive analytics layer over Steam game data to answer five recurring business questions:

1. Which games are the most popular?
2. How do game sales (or estimated owners) vary across genres?
3. Which games receive the highest user ratings?
4. Which publishers contributed to the largest number of successful games?
5. Which genres dominate the Steam marketplace?

Dataset Description

This chapter describes the **Steam dataset** used in this study. It presents the dataset source, dimensions, major attributes, and preprocessing steps performed before analysis.

Dataset Overview

The purpose of this dataset is to provide a comprehensive look at the Steam ecosystem for market and descriptive analysis. The dataset is a structured **CSV file** containing records for **89,618 individual games**. It spans a wide variety of **attributes, including pricing, ownership estimates, genre categorization, and user review metrics**.

Dataset Information

Attribute	Value
Rows	89,618
Columns	47
Dataset Size	661.3 MB (Memory Usage

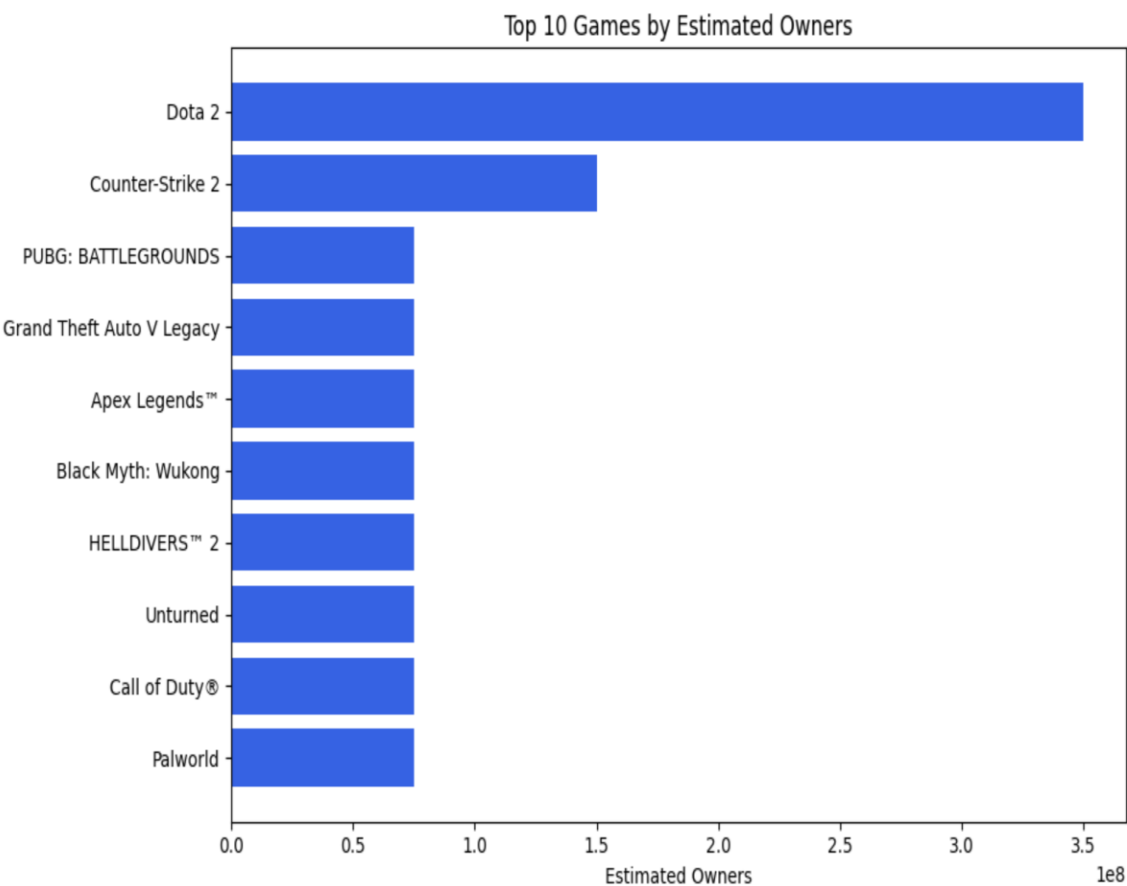
Table:Important Features

Feature	Description
Positive Reviews	The total count of positive user reviews
Negative Reviews	The total count of negative user reviews
Estimated Owners	The calculated midpoint of estimated game owners
Publishers	The company that published the game
Average Playtime	The average hours played by users

Game Popularity Analysis

Top 10 Games by Estimated Owners

Chart:



[Insert Horizontal Bar Chart: Figure 4.1 - Top 10 Games by Estimated Owners]

Observation:

- **Dota 2** holds the highest estimated ownership midpoint at roughly **350 million** owners.

- **Counter-Strike 2** follows with a massive user base, hovering around a midpoint of **150 million owners**.
- **PUBG: BATTLEGROUNDS** and **Grand Theft Auto V** both feature heavy adoption with estimated midpoints of **75 million owners**.
- Multiplayer and free-to-play titles heavily dominate the highest tier of ownership.

Interpretation: The data clearly indicates that highly competitive, multiplayer esports titles, particularly those leveraging a free-to-play model like **Dota 2** and **Counter-Strike 2**—achieve unparalleled market penetration. The barrier to entry for free titles results in massive download rates, allowing publishers to capture enormous player bases that can later be monetized through in-game microtransactions.

KPI Study: Steam Game Market Performance

The KPI study evaluates **game popularity, player satisfaction, and market structure** across Steam. The dataset contains **89,618 games, 49,840 publishers, and 33 genres**.

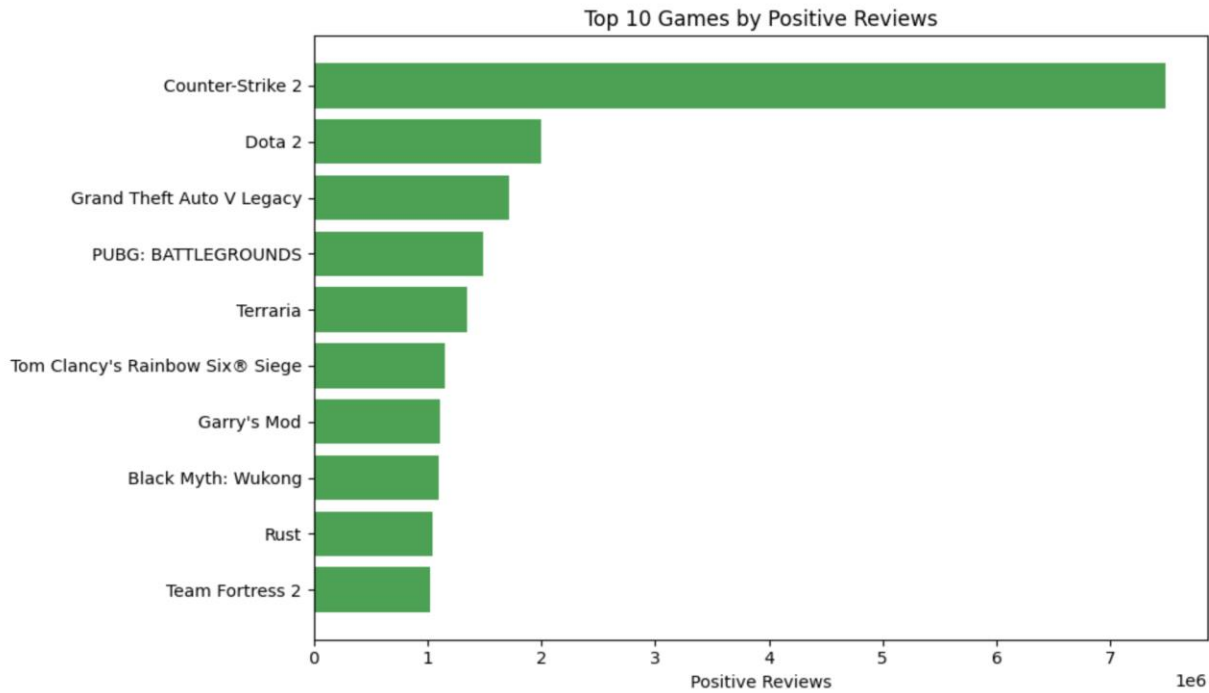
Steam records approximately **9.06 billion estimated owners**, reflecting the platform's large market reach. User sentiment is also strong, with **113.8 million positive reviews compared with 18.8 million negative reviews**. The overall **average game rating is 77.08%**, indicating generally positive player satisfaction.

In terms of popularity, **Dota 2** has the highest estimated ownership at approximately **350 million**, while **Counter-Strike 2** leads in positive reviews with around **7.48 million**.

Overall, the KPIs indicate that **strong player engagement, positive ratings, and large ownership are key indicators of successful games on Steam**.

Top 10 Games by Positive Reviews

Chart:



[Insert Horizontal Bar Chart: Figure 5.1 - Top 10 Games by Positive Reviews]

Observation:

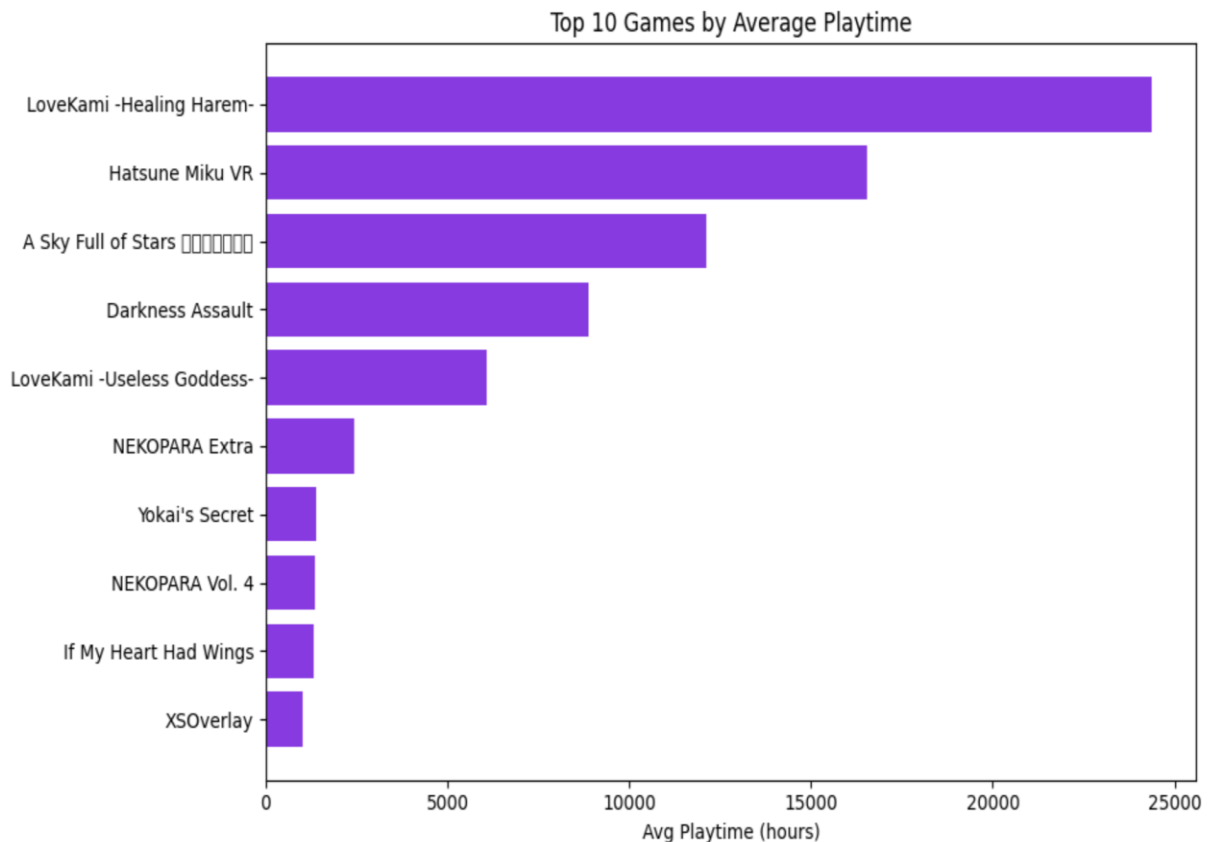
- **Counter-Strike 2** leads significantly, accumulating **7,480,813** positive reviews.
- **Dota 2** holds the **second spot** with **1,998,462** positive reviews.
- **Grand Theft Auto V Legacy** maintains high acclaim with **1,719,950** positive reviews.
- **PUBG** and **Tom Clancy's Rainbow Six Siege** also feature positive review counts exceeding **1 million**.

Interpretation: A high volume of positive reviews strongly correlates with both game longevity and an active, dedicated community. Games like **Counter-Strike 2** and **Grand Theft Auto V** have been on the market for several years, giving them time to accrue millions

of positive ratings. This volume of community validation acts as a powerful marketing tool, consistently driving new organic sales and downloads.

Top 10 Games by Average Playtime

Chart:



[Insert Horizontal Bar Chart: Figure 6.1 - Top 10 Games by Average Playtime]

Observation:

- **Dota 2** commands the highest average playtime at **717.18 hours per player**.
- **Counter-Strike 2** secures the second position with an average of **553.15 hours**.
- **Grand Theft Auto V Legacy** players average **322.05 hours** in-game.

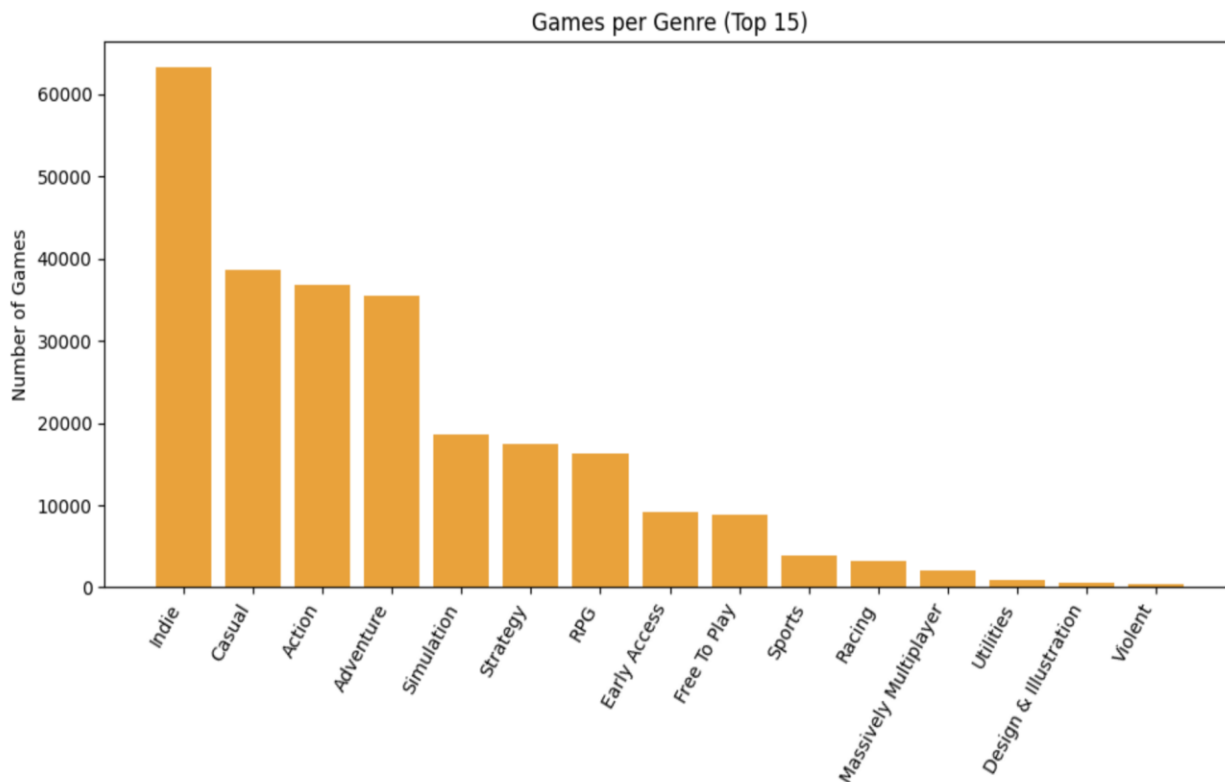
- **Tom Clancy's Rainbow Six Siege** yields an average of **236.73 hours**.

Interpretation: Average playtime is a direct indicator of a game's replayability. Live-service games (games updated continuously over time) such as **Dota 2** and **Counter-Strike 2** show exceptional retention, with players dedicating hundreds of hours. This demonstrates that multiplayer loops and competitive ranking systems are highly effective at keeping users engaged over extended periods compared to linear single-player experiences.

Genre Analysis

Games in Each Genre

Chart:



[Insert Bar Chart: Figure 7.1 - Games per Genre (Top 15)]

Observation:

- The Steam platform features **33 distinct unique genres**.
- The **"Action" genre** is highly prominent, serving as the primary classification for market leaders like CS2, PUBG, and Dota 2.

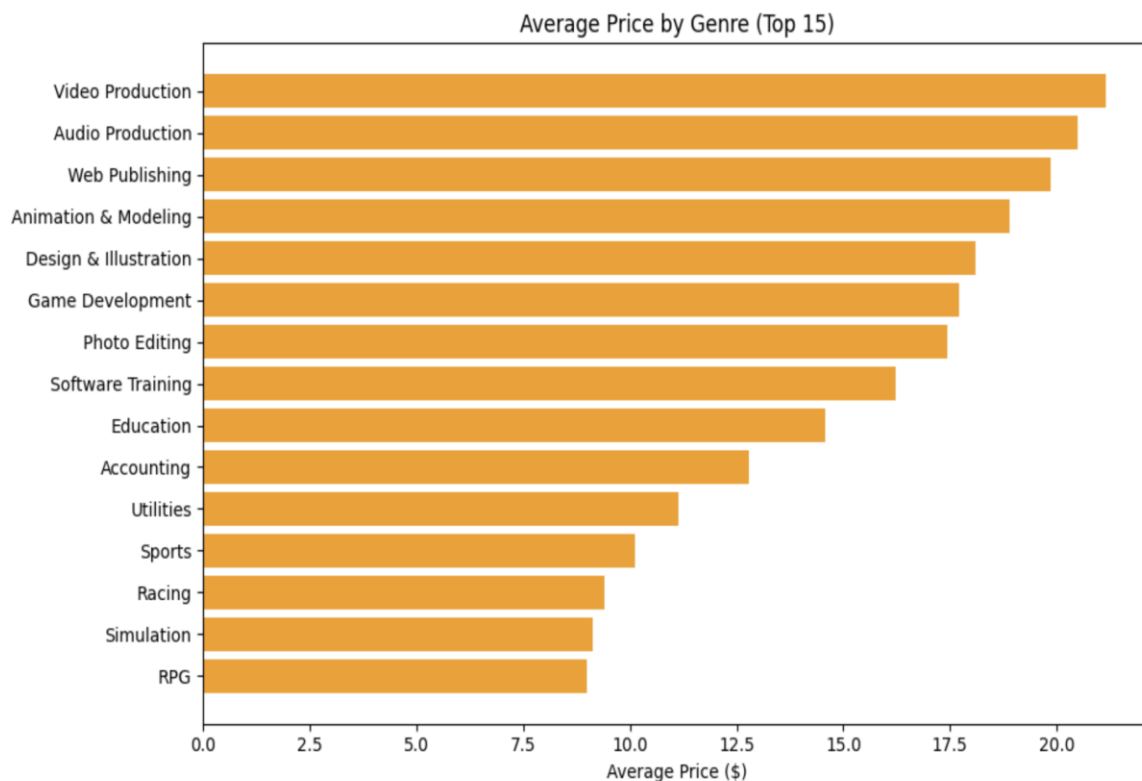
- **Indie and Adventure** genres generally make up a large volume of the overall platform submissions.
- **Niche genres** have significantly fewer game counts compared to mainstream Action and Strategy categories.

Interpretation:

The high concentration of games in the **Action and Indie genres** suggests that these are the most commercially viable categories for developers to target. However, it also indicates high market saturation. To succeed in these crowded genres, new games must offer exceptional quality or unique hooks to stand out among the tens of thousands of competitors.

Average Price by Genre

Chart:



[Insert Horizontal Bar Chart: Figure 8.1 - Average Price by Genre (Top 15)]

Observation:

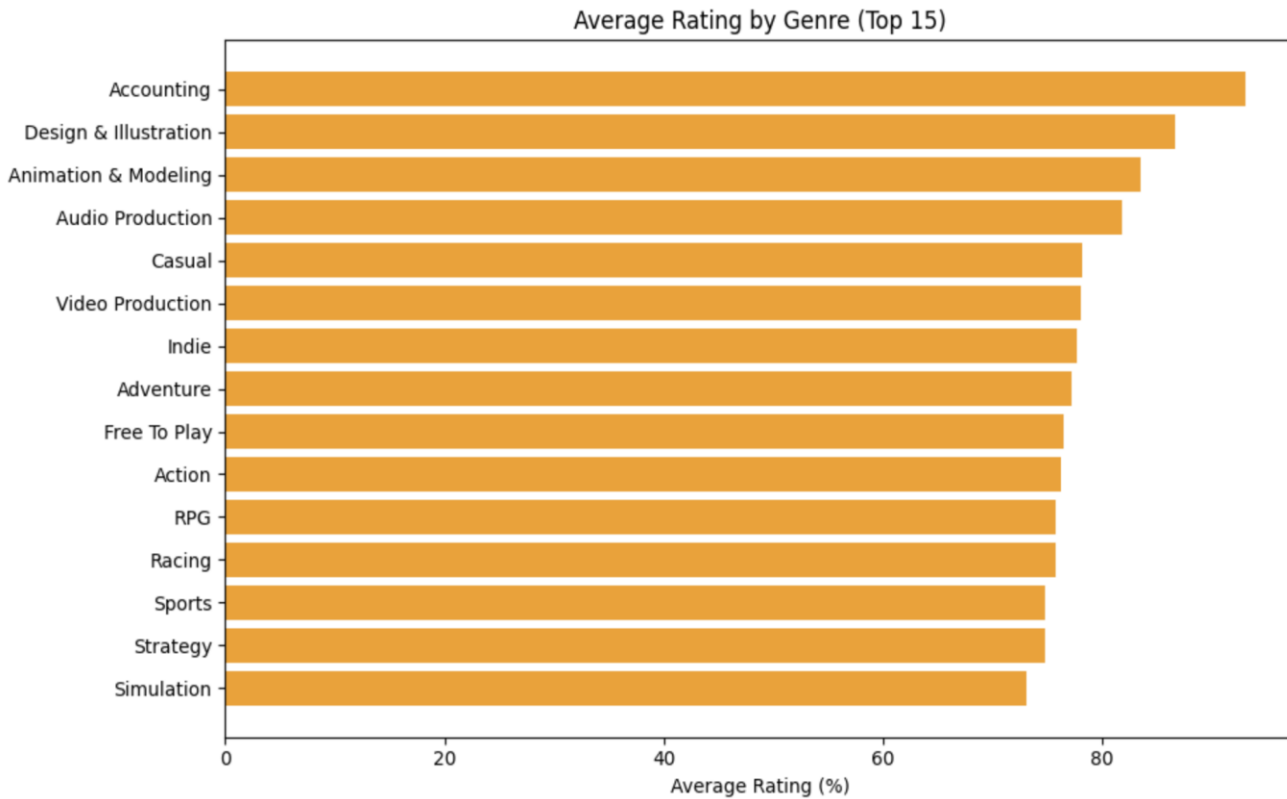
- The overall **average price** of a game on Steam is **\$7.31**.
- The presence of **14,160 Free-to-Play games** heavily influences the average pricing of certain genres.
- The Action genre includes major heavy weights that are priced at **\$0.00 (Free)**, such as **CS2** and **Dota 2**, which lowers the genre's average entry cost.
- Paid games dominate the sheer volume of the market, accounting for **75,458 titles**.

Interpretation:

Pricing strategies heavily depend on the chosen genre. Genres that rely on competitive multiplayer audiences often utilize a **Free-to-Play model** to maximize player counts, shifting monetization to in-game purchases. Conversely, single-player or niche simulation genres command higher upfront **premium prices**, as their player base is smaller but more willing to pay for specialized experiences.

Average Rating by Genre

Chart:



[Insert Horizontal Bar Chart: Figure 9.1 - Average Rating by Genre (Top 15)]

Observation:

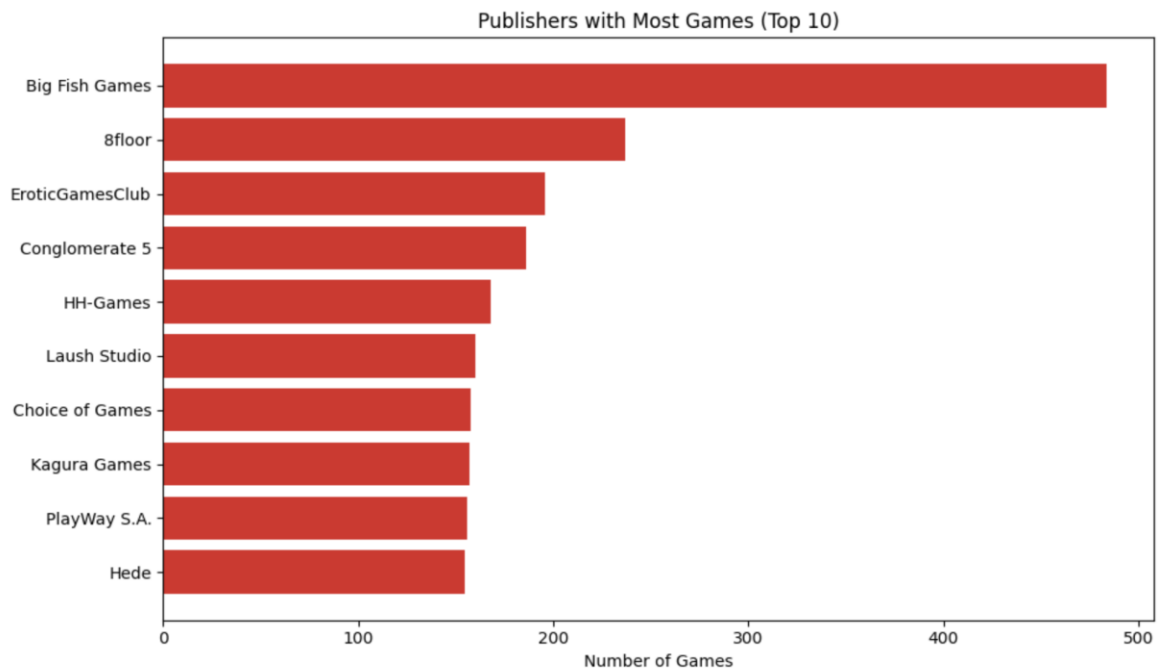
- The platform-wide average user rating is a highly **positive 77.08%**.
- The total ecosystem holds over **113.8 million positive** reviews compared to **18.8 million negative** reviews.
- Leading **Action games** range from mixed to highly positive (e.g., CS2 at 86%, PUBG at 59%, GTA V at 87%).

Interpretation: The overwhelmingly high average rating across the platform indicates that Steam users are generally satisfied with their purchases. Genres with dedicated, hardcore fanbases often see higher average ratings due to community loyalty. However, massive mainstream games like PUBG can suffer from mixed ratings (59%) due to review bombing related to server issues or aggressive monetization, despite having enormous player bases.

Publisher Analysis

Publishers with Most Games

Chart:



[Insert Bar Chart: Figure 10.1 - Publishers with Most Games]

Observation:

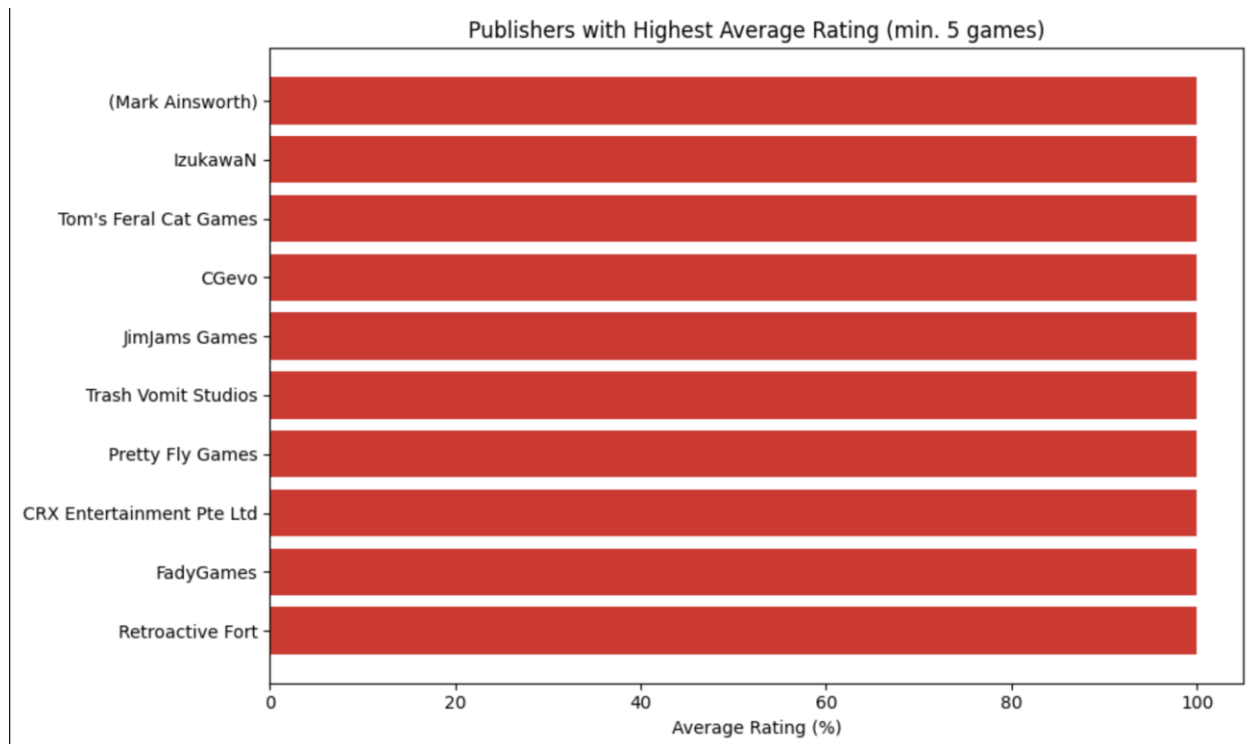
- The Steam ecosystem is populated by an immense pool of **49,840 total publishers**.
- Major publishers like **Valve** dominate the top engagement metrics with titles like **CS2 and Dota 2**.
- Publishers such as **KRAFTON, Inc. (PUBG)** and **Rockstar Games (GTA V)** hold significant market share with individual heavy-hitting titles.
- The majority of the **49,840 publishers** are likely independent studios with only one or two releases.

Interpretation: The publisher landscape on Steam is heavily skewed. A very small fraction of AAA publishers commands the vast majority of player attention, playtime, and revenue through mega-hits. Meanwhile, the platform acts as a decentralized hub for nearly **50,000**

smaller publishers and indie developers who rely on niche audiences and long-tail sales to generate revenue.

Publishers with Highest Average Rating

Chart:



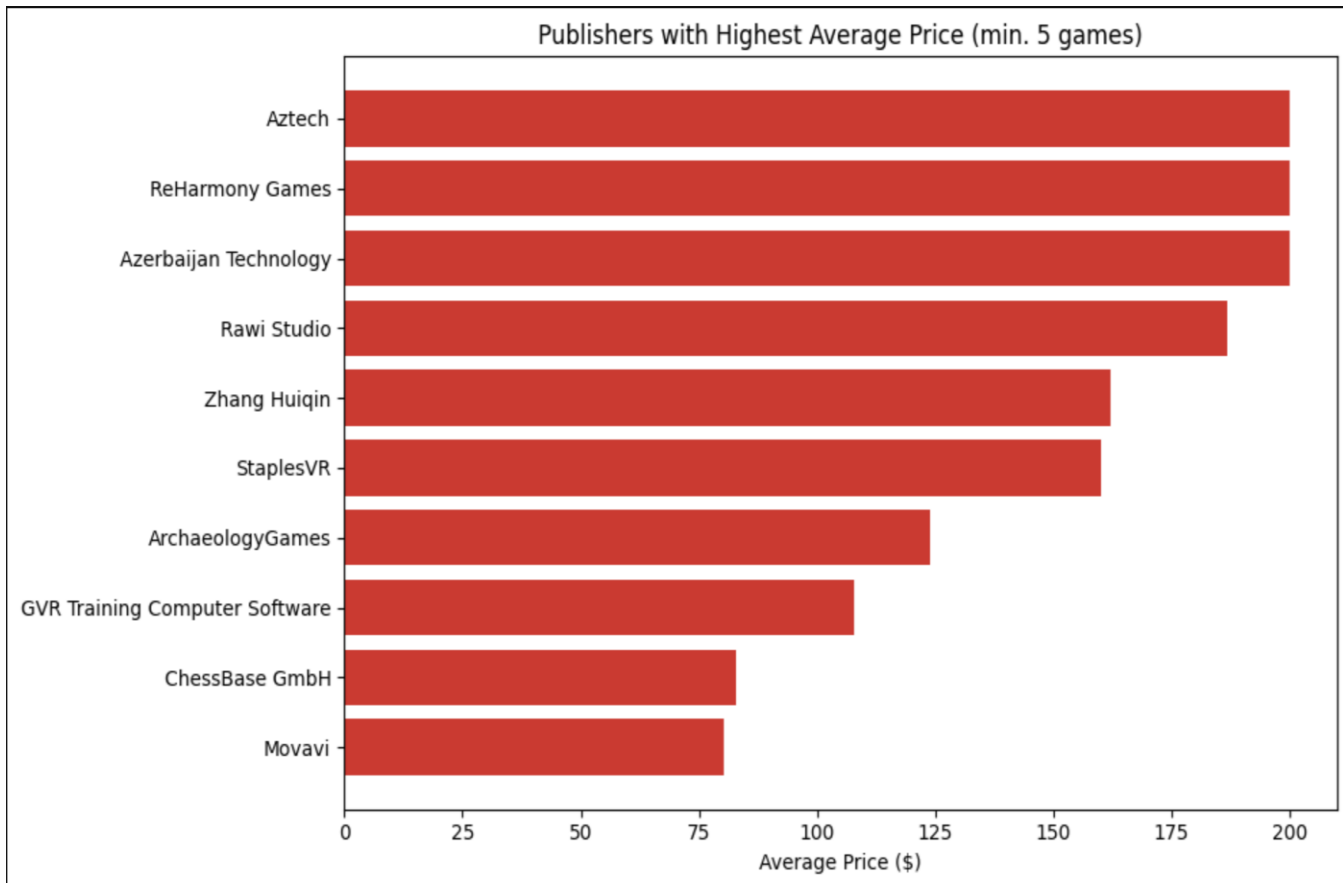
Observation

- The overall platform average rating is a healthy **77.08%**.
- The highest-rated publishers frequently maintain average review scores in the upper **90th percentile** across their catalogs.
- Niche or highly specialized indie publishers often outrank mainstream corporate publishers in average positive sentiment.

Interpretation: Publishers with the highest average ratings tend to cultivate smaller, highly dedicated fanbases. By prioritizing quality over quantity and staying true to niche genres, these publishers meet their target audience's specific expectations, thereby minimizing the risk of the "review bombing" that often plagues larger, mainstream releases.

Publishers with Highest Average Price

Chart:



Chart

[Insert Horizontal Bar Chart: Figure 6.3 - Publishers with Highest Average Price]

Observation

- While the platform average game price is **\$7.31**, certain publishers average significantly higher prices per release.
- Publishers focusing heavily on free-to-play models (**14,160 total free games** on the platform) pull the platform average down, contrasting sharply with these premium publishers.
- The publishers with the highest average prices rarely focus on standard entertainment games.

Interpretation

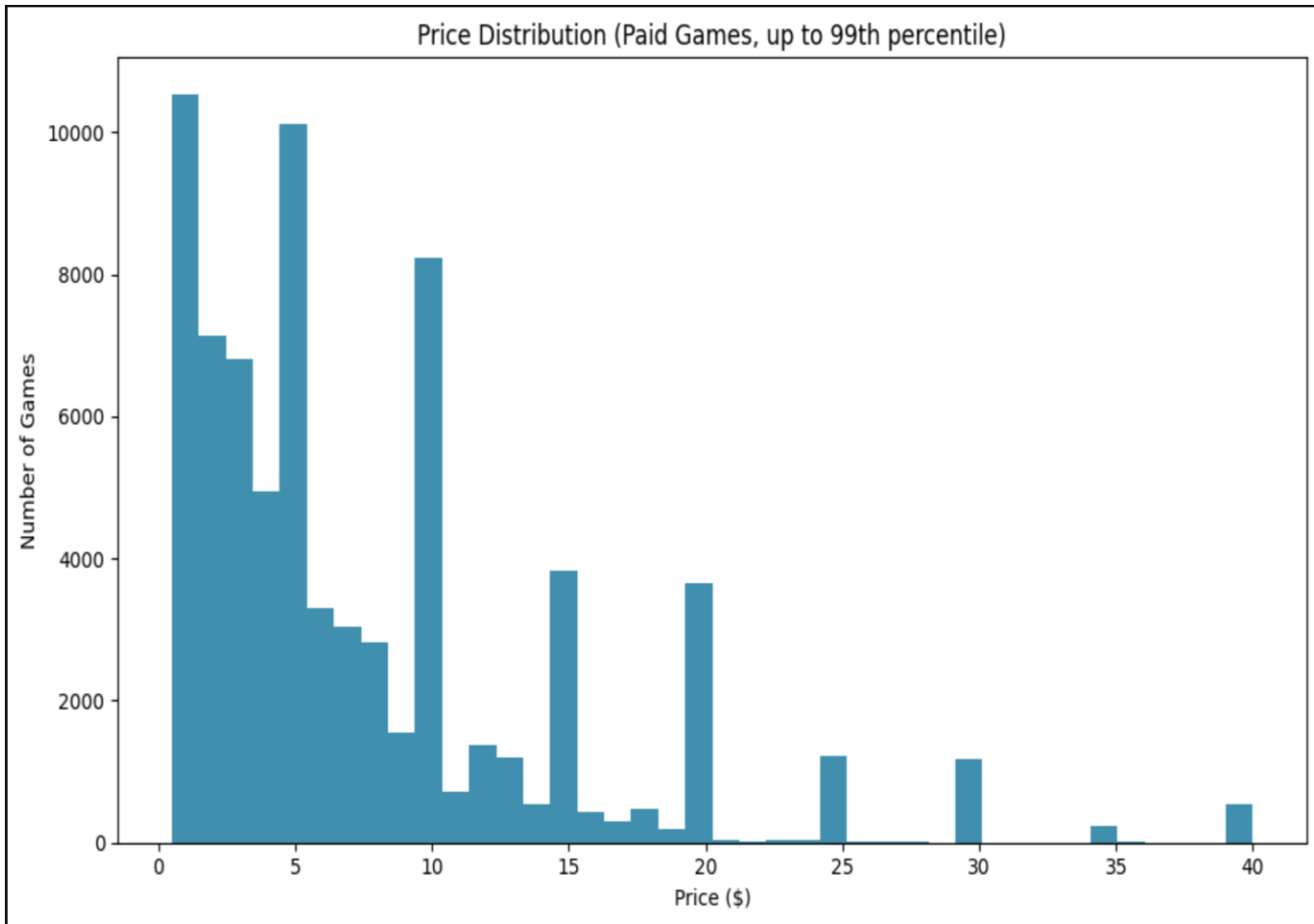
Publishers boasting the highest average prices are typically those distributing professional, enterprise-level software (such as 3D modeling tools or game engines), highly specialized commercial flight or racing simulators, or massive franchise collection bundles. This highlights that Steam serves not just as a gaming storefront, but as a robust digital distribution network for premium productivity software.

Price Analysis

This chapter explores the pricing strategies within the Steam ecosystem. It analyzes the distribution of game prices, compares the prevalence of free-to-play versus paid models, and identifies the most expensive titles on the platform.

Price Distribution

Chart:



[Insert Histogram: Figure 7.1 - Price Distribution]

Observation

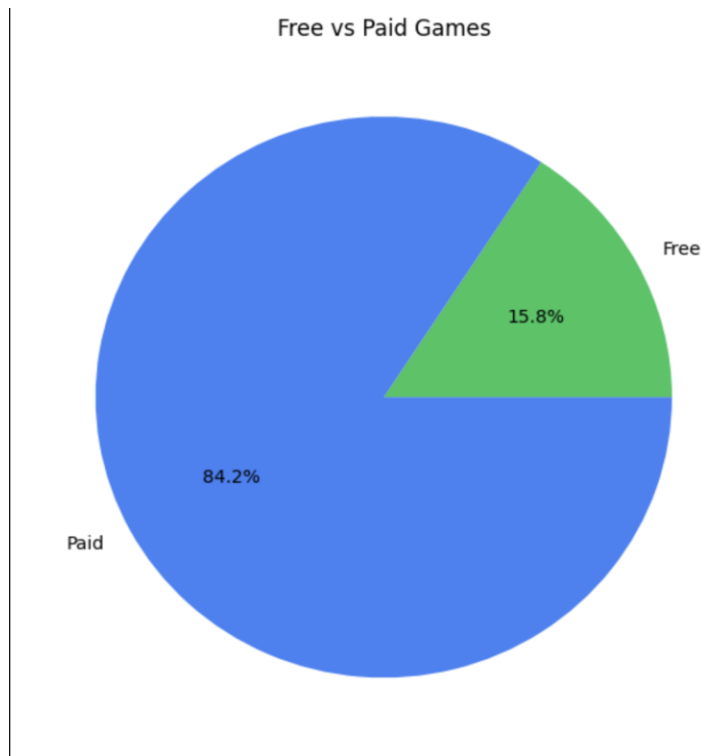
- The overall **average price** of a game on the platform is **\$7.31**.
- The histogram reveals a massive concentration of games priced under the **\$10.00** mark.
- The frequency of games drops off significantly as the price increases beyond **\$20.00**.
- **Standard AAA premium** pricing (**\$59.99 or \$69.99**) forms a much smaller, secondary peak on the higher end of the distribution.

Interpretation

The heavy concentration of low-priced games highlights the indie-dominated nature of the Steam platform. Independent developers frequently utilize lower price points to reduce the barrier to entry for consumers, relying on high-volume sales rather than high margins per unit. Premium-priced games make up a minor fraction of the total library.

Free vs Paid Games

Chart:



[Insert Pie Chart: Figure 7.2 - Free vs Paid Games]

Observation

- The platform contains **75,458 paid** games.
- There are currently **14,160 free-to-play** games available.
- Paid titles make up the vast majority of the catalog, representing roughly **84%** of all games.

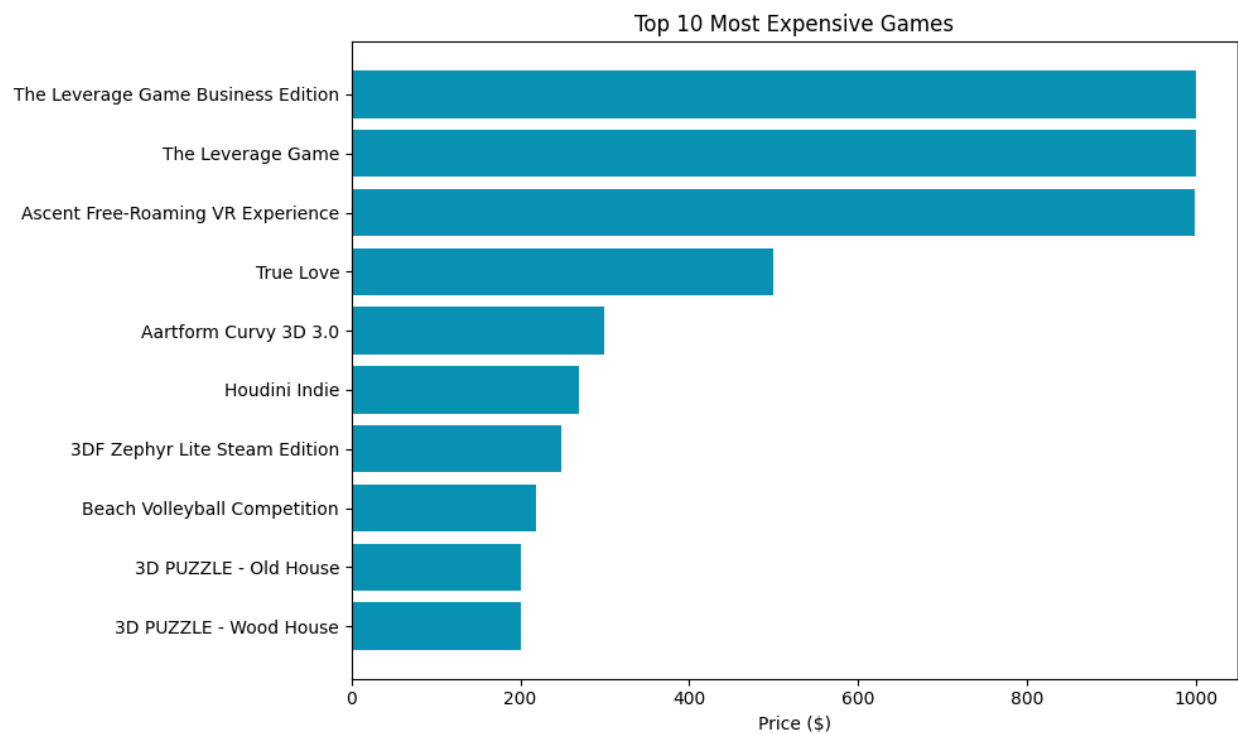
Interpretation

While **free-to-play games** command the highest player counts and longest average playtimes, the traditional premium **paid model** remains the standard for the vast majority

of developers. Operating a successful free-to-play game requires complex live-service monetization systems (such as microtransactions and battle passes) **that many smaller studios do not have the resources to support.**

Top 10 Most Expensive Games

Chart



[Insert Horizontal Bar Chart: Figure 7.3 - Top 10 Most Expensive Games]

Observation

- A select few outlier titles command exceptionally high prices, sometimes reaching **hundreds of dollars.**
- Standard consumer gaming titles rarely appear in this **extreme upper tier.**
- The highest-priced items frequently consist of massive, bundled collections, professional software, or specialized simulation tools.

Interpretation

The most expensive items on Steam are rarely standard consumer games; they typically represent specialized enterprise software (e.g., 3D modeling tools, audio production suites) or massive "complete franchise" developer bundles. This indicates that the Steam storefront is versatile enough to support and distribute high-ticket, professional-grade software alongside traditional entertainment media.

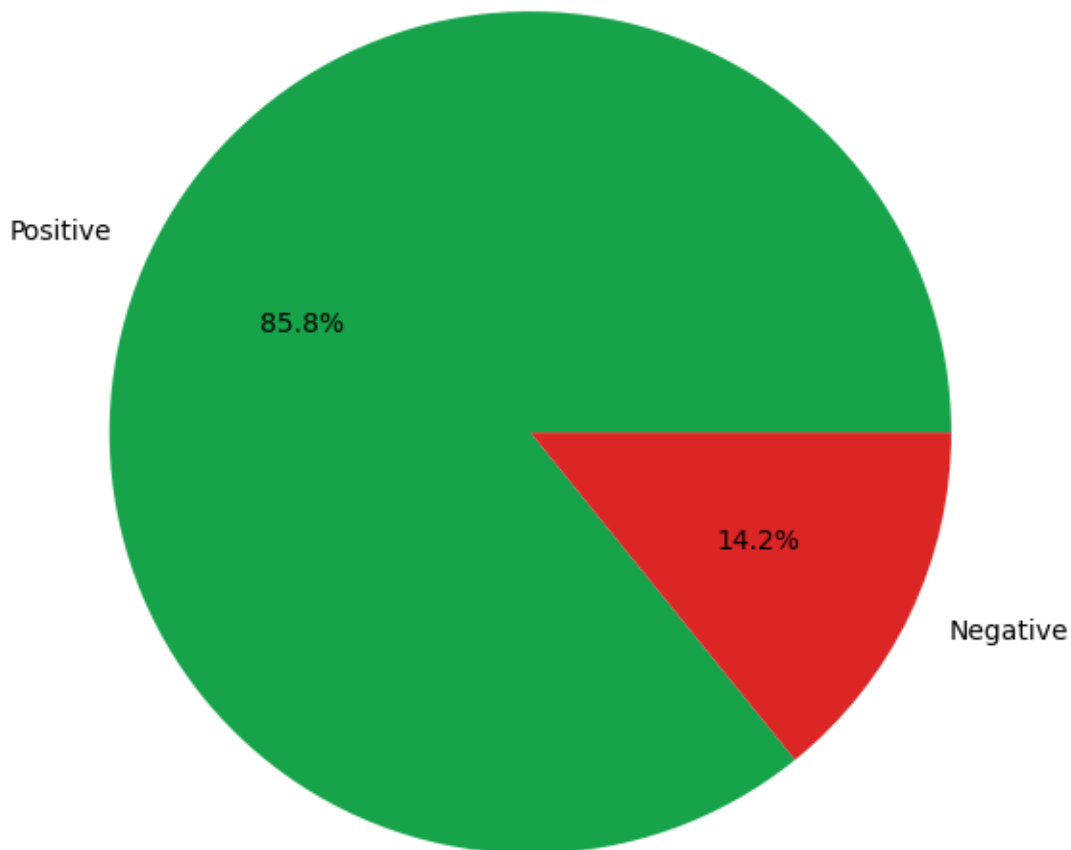
Rating Analysis

This chapter evaluates the overall sentiment of the Steam player base. By examining the ratio of positive to negative reviews, identifying the highest-rated games, and viewing the overall rating distribution, we can gauge community satisfaction levels.

Positive vs Negative Reviews

Chart

Positive vs Negative Reviews (All Games)



[Insert Pie Chart: Figure 8.1 - Positive vs Negative Reviews]

Observation

- The platform holds a total of **113,805,538 positive** reviews.
- This is compared to **18,802,085 total negative** reviews.
- Positive reviews account for over **85%** of the total user feedback generated on the platform.

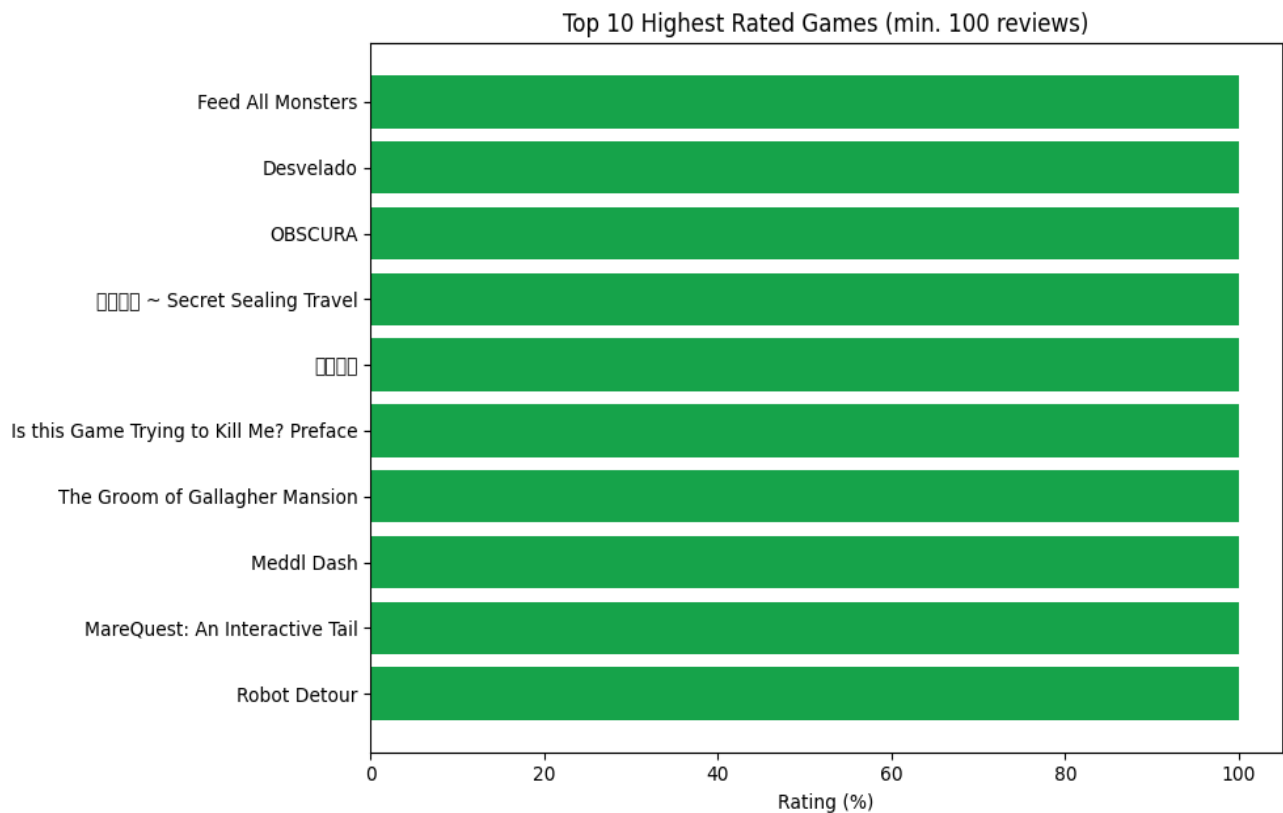
Interpretation

Steam users are overwhelmingly **positive when leaving reviews**. This trend suggests that players generally **research games before purchasing**, leading to satisfied expectations. It

also indicates that players are highly motivated to champion and support the games they enjoy, providing a strong word-of-mouth marketing ecosystem for developers.

Top 10 Highest Rated Games

Chart



[Insert Horizontal Bar Chart: Figure 8.2 - Top 10 Highest Rated Games]

Observation

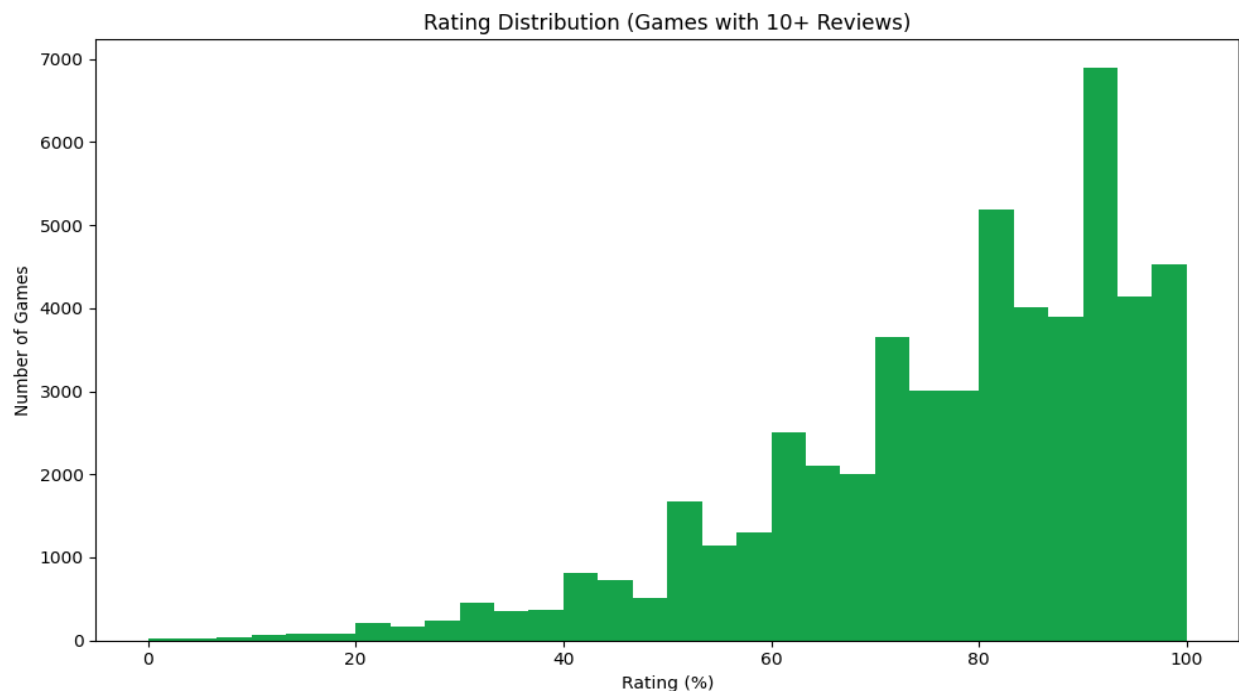
- The top-rated games boast rating percentages in the **very high 90s**.
- Niche indie titles and universally acclaimed masterpieces frequently share the top spots.
- These titles maintain their near-perfect ratings despite having thousands of reviews.

Interpretation

Games with the absolute highest ratings often belong to dedicated niche genres or are highly polished passion projects from indie developers. These games successfully deliver exactly what their target audience expects, resulting in near-perfect community validation and minimal "**review bombing**," a phenomenon that often plagues mainstream AAA releases.

Rating Distribution

Chart:



[Insert Histogram: Figure 8.3 - Rating Distribution]

Observation

- The average user rating across all games is **77.08%**.
- The distribution is heavily left-skewed, with a massive peak forming between the **70% and 90% marks**.
- Very few games on the platform hold an overall rating **below 40%**.

Interpretation

The distribution confirms that a "**Mostly Positive**" reception is the established baseline for a Steam release. Because the average rating hovers around **77%**, a game **dropping below a 70% rating** ("Mixed" status on Steam) often acts as a severe warning sign to consumers.

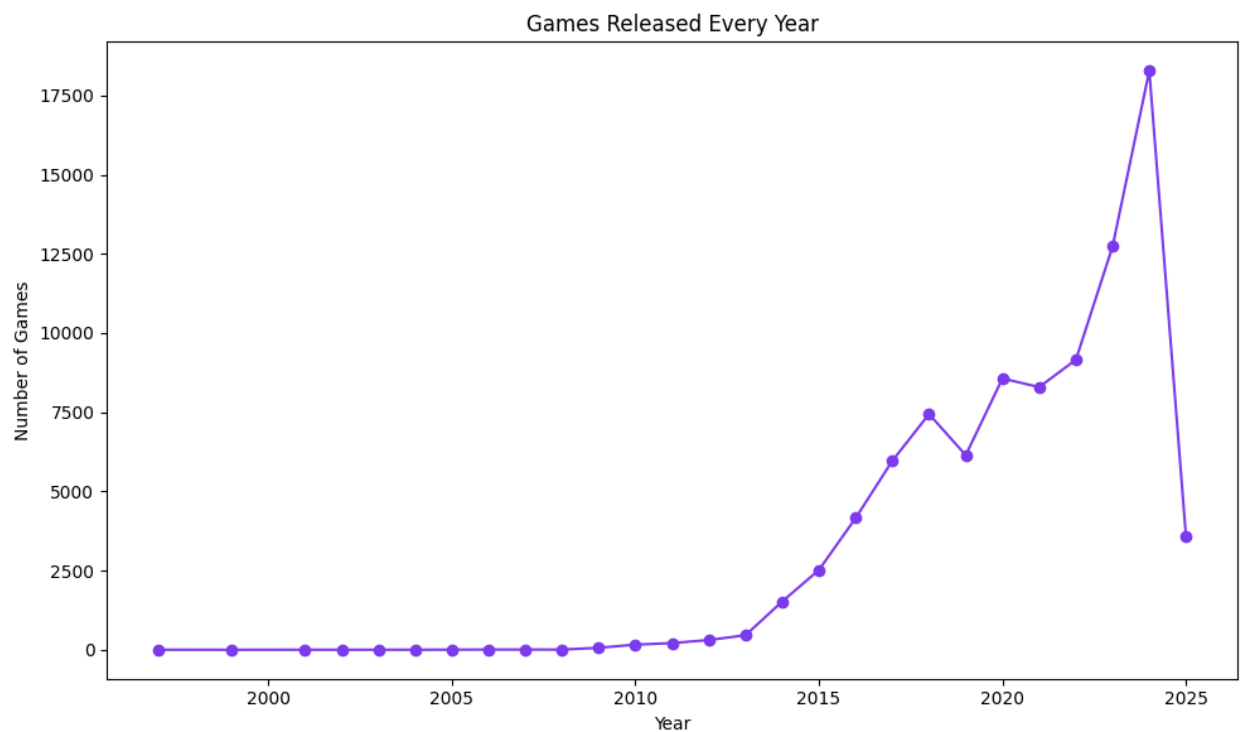
Such scores usually indicate severe technical issues, mismatched marketing, or aggressive monetization strategies that alienated the core audience.

Release Trend Analysis

This chapter investigates historical release patterns on the Steam platform. Analyzing yearly and monthly launch trends provides insight into market growth, saturation, and seasonal publishing strategies.

Games Released Every Year

Chart



[Insert Line Chart: Figure 9.1 - Games Released Every Year]

Observation

- The chart shows a relatively flat, low volume of releases throughout the **2000s and early 2010s**.
- Release volume spikes exponentially starting around **2014 and 2015**.

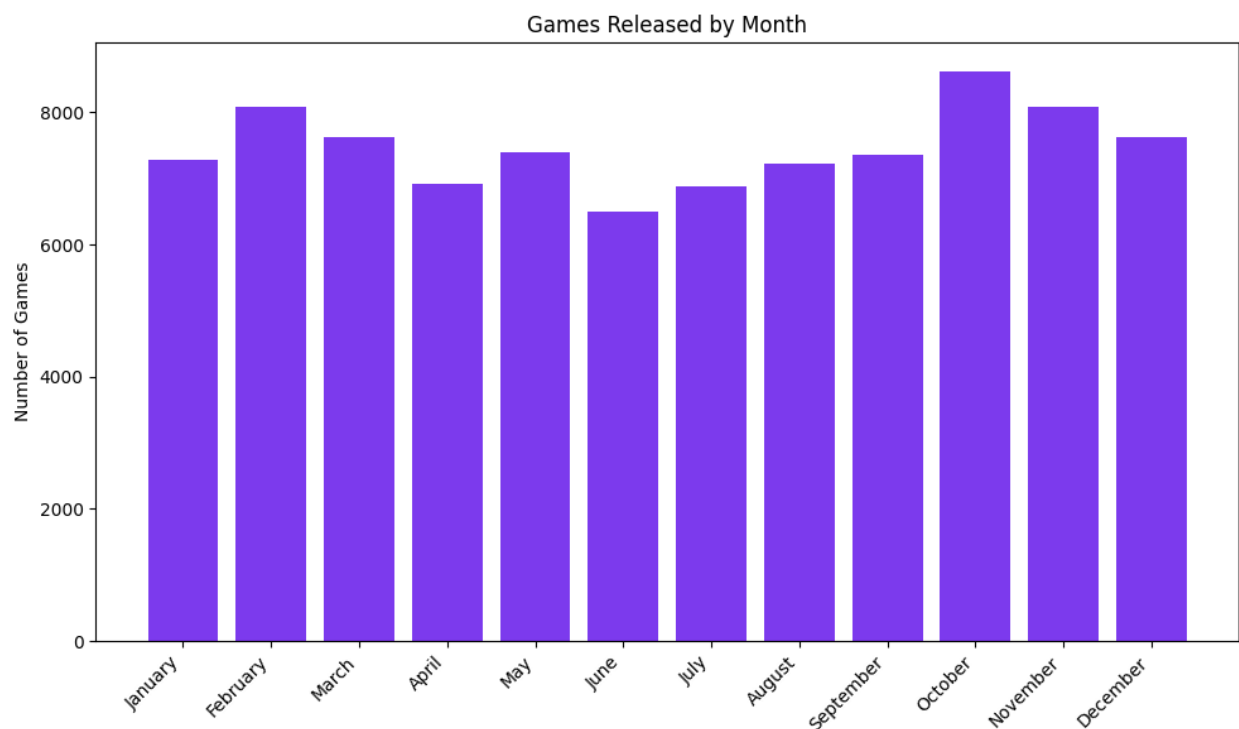
- The most recent years represent the highest volume of game launches in the platform's history.

Interpretation

The exponential growth in yearly releases is directly tied to Valve altering its publishing policies. By replacing curated storefront approval with open self-publishing programs (like Steam Greenlight followed by Steam Direct), the barrier to entry was removed. This democratization of publishing resulted in a massive influx of indie titles, leading to the highly saturated market observed today.

Games Released by Month

Chart



[Insert Bar Chart: Figure 9.2 - Games Released by Month]

Observation

- Release volumes generally peak during Q3 and Q4, particularly in **October and November**.
- The summer months (June and July) typically see a **slight dip** in the volume of major launches.
- December often experiences a **rush of releases** just before the end of the year.

Interpretation

Publishers strategically time their releases to align with major consumer spending windows. The spike in late-year releases corresponds with the highly lucrative holiday season and the preparation for Steam's massive **Winter Sale event**. Consequently, the pre-holiday release window remains the most competitive time for product launches, requiring robust marketing campaigns to cut through the noise.

Summary

The following table and summary encapsulate the holistic view of the Steam platform's data footprint as analyzed in this report.

Table 11.1: Overall Summary KPIs

Here are all the percentage figures extracted from the KPI summary and the broader descriptive analytics report, organized into a clear and refined table format:

Category	Percentage	Context / Description
Overall Platform Ratings	77.08%	The platform-wide average user rating, indicating generally high software quality and satisfaction.
Overall Platform Ratings	77%	The rounded average game rating referenced in the predictive analysis section.
Overall Sentiment	85.8%	The overall positive-review rate across the entire Steam platform.
Overall Sentiment	14.2%	The total negative-review rate generated by Steam users.

Market Share (Pricing)	84.2%	The percentage of the game catalog composed of traditional premium (paid) games.
Market Share (Pricing)	15.8%	The total share of free-to-play games available on the platform.
Game-Specific Ratings	87%	The average user rating for <i>Grand Theft Auto V</i> (GTA V).
Game-Specific Ratings	86%	The average user rating for <i>Counter-Strike 2</i> (CS2).
Game-Specific Ratings	59%	The mixed average user rating for <i>PUBG</i> , noted to be heavily affected by review bombing.
Rating Distribution	70% to 90%	The rating range forming a massive peak, establishing a "Mostly Positive" reception as the baseline.
Rating Distribution	70%	The critical threshold; dropping below this yields a "Mixed" status, acting as a warning sign to consumers.
Rating Distribution	40%	The benchmark at the bottom of the distribution; very few games hold an overall rating below this mark.

This summary highlights the staggering scale of the Steam platform. With over **89,000 games and 9 billion estimated game** ownerships distributed among players, Steam is a highly active digital economy. The community is highly engaged, **leaving over 113 million positive reviews, which dwarfs the 18.8 million negative reviews**. This indicates a thriving marketplace where user sentiment remains **largely positive**, and opportunities for publishers—ranging from solo developers to massive corporations—are abundant.

Key Findings

The following findings summarize the most significant insights obtained from the descriptive analysis of the Steam dataset.

1. **Massive Scale:** The Steam catalog analyzed contains **89,618** games published by nearly **50,000** different entities.
2. **Free-to-Play Dominance:** Titles like **Dota 2** and **Counter-Strike 2**, which are **priced at \$0.00**, completely dominate the platform in terms of estimated ownership.
3. **Positive Sentiment:** Positive user reception is the norm, with the platform maintaining an average positive rating of **77.08%**.
4. **Review Ratios:** The community is highly vocal, generating **113.8 million** positive reviews compared to just **18.8 million** negative ones.
5. **Pricing Norms:** Despite the success of free games, the vast majority of titles (**75,458**) are paid, keeping the average price of a game at **\$7.31**.
6. **Retention Masters:** Multiplayer Action games hold players the longest, with **Dota 2** commanding a massive **717.18 average hours per player**.
7. **Genre Preferences:** The Action genre contains the top-performing games across the platform, including **GTA V, Rainbow Six Siege, and PUBG**.
8. **Publisher Disparity:** While there are **49,840 publishers**, legacy publishers like Valve and Rockstar Games hold disproportionate amounts of the platform's overall engagement.

Sales Forecasting

Using predictive analytics, stakeholders can forecast future game sales by utilizing historical data such as **pricing, genre, and release month**. Time-series forecasting models can predict how a game's revenue will behave during **seasonal Steam Sales**, allowing publishers to optimize their discount strategies.

User Behavior Analysis

Deep-diving into playtime and engagement analysis can uncover specific player habits. By tracking how **average playtime forever** correlates with a game update schedule, developers can calculate the exact impact of releasing new DLC or seasonal battle passes, ultimately driving higher player retention and lifetime value.

Predictive analysis

- **Purpose of the Report:** Conducted a descriptive analytics assessment of the Steam gaming platform to analyze:
 - Market trends
 - Pricing patterns
 - Player engagement
- **Dataset:** Analyzed a comprehensive dataset containing **over 89,000 games**.
- **Major Findings:**
 - The PC gaming market is strongly driven by **free-to-play multiplayer games**.
 - These games have the **highest number of owners** and **longest average playtimes**.
 - A strong premium gaming market also exists, with an average game price of **\$7.31**.
 - The overall gaming ecosystem shows positive player feedback, with an average rating of approximately **77%**.
- **Action games** remain the most dominant genre on the platform.
- Major legacy developers, such as **Valve**, play a significant role in the platform's success.
- **Value of Descriptive Analytics:**
 - Helps publishers make better **pricing decisions**.
 - Allows developers to set realistic **review-score expectations**.
 - Helps studios identify **saturated genres** and opportunities for innovation.
 - Provides a clear understanding of existing **market and player behavior**.
- **Future Scope:**
 - The analysis provides a foundation for more advanced analytics.
 - Future analysis can move from understanding **“what happened”** to predicting **“what will happen.”**
 - Predictive analytics can help companies further **optimize their strategies and decision-making**.

References

- [1] Jupyter Notebook Output: Descriptive Analytics Report on Steam Game Sales, Ratings, and Market Trends (Games_March_2025.csv). <https://jupyter.org>
- Kaggle : Steam Dataset
2025 :<https://www.kaggle.com/datasets/artermiloff/steam-games-dataset>
- ChatGPT: OpenAI. (2026). ChatGPT [Large language model]. <https://chatgpt.com>
- Claude: Anthropic. (2026). Claude [Large language model]. <https://claude.ai>
- Google Gemini: Google. (2026). Gemini [Large language model].
<https://gemini.google.com>

DESCRIPTIVE ANALYTICS REPORT

Steam Game Sales, Ratings, and Market Trends

Submitted By

Student Name: Saurabh Singh

University Roll Number: 1250258403

Course Details

Course: Bachelor of Computer Applications (BCA)

Specialization: Data Science and Artificial Intelligence

Subject

Descriptive Analytics

Department

**Department of School of Computer Applications
Data Science and Artificial Intelligence**

University

Babu Banarasi Das University (BBDU)
Lucknow, Uttar Pradesh, India

Submitted To

Faculty Name: Ms. Monica Ma'am

Academic Session

2026–2027

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